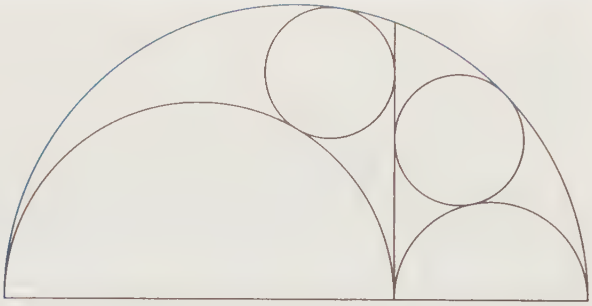


Archimedes' Circles

Let no one ignorant of geometry enter here. – Inscription above Plato's Academy

CHAPTER 19 Problem Solving Strategies in Geometry

In this chapter we review many of the most powerful geometry problem-solving strategies we have learned in the text by tackling some challenging problems with them. We also apply many of our geometric tools to proofs that require multiple insights.

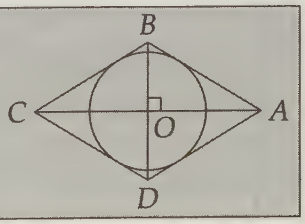
This chapter is meant to extend the lessons of this book, so many of the Exercises and Challenge Problems in this chapter are significantly more difficult than most of the problems elsewhere in this text. One excellent strategy for mastering the problems you cannot solve on your first try is to review the solutions, then try the problems again on your own a few days later.

19.1 The Extra Line

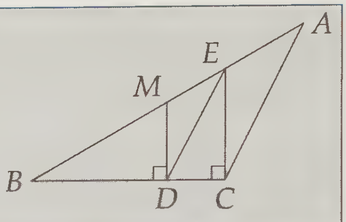
We cleverly added extra lines to diagrams to solve various problems throughout this book. In this section we reinforce the most common indications that extra lines might be helpful.

Problems

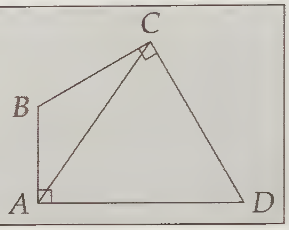
Problem 19.1: Find the radius of a circle that is inscribed in a rhombus that has diagonals of length 16 and 30.



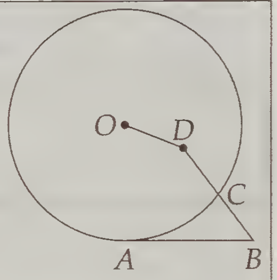
Problem 19.2: In obtuse triangle ABC shown, $AM = MB$, $\overline{MD} \perp \overline{BC}$, and $\overline{EC} \perp \overline{BC}$. If the area of $\triangle ABC$ is 24, what is the area of $\triangle BED$? *(Source: AMC 12)* **Hints:** 27, 541



Problem 19.3: In triangle ABC, $\angle ABC = 120°$, $AB = 3$ and $BC = 4$. If lines perpendicular to $\overline{AB}$ at A and to $\overline{BC}$ at C meet at D, then find CD. *(Source: AMC 12)*

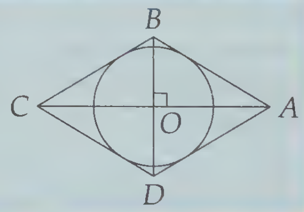


Problem 19.4: $\overline{AB}$ is tangent at A to the circle with center O, point D is interior to the circle, and $\overline{DB}$ intersects the circle at C. If $BC = CD = 3$, $OD = 2$, and $AB = 6$, then find the radius of the circle. *(Source: AMC 12)*

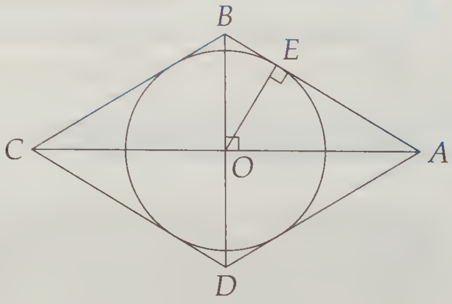


One very common type of 'extra line' we draw is a perpendicular from a point to a line. Usually, our goal in doing so is to build a useful right triangle. Here's a classic example.

Problem 19.1: Find the radius of a circle that is inscribed in a rhombus that has diagonals of length 16 and 30.



Solution for Problem 19.1: Circles + tangent lines = draw a radius to a point of tangency. We draw radius $\overline{OE}$ because $\overline{OE} \perp \overline{AB}$. Since the diagonals of a rhombus are perpendicular and bisect each other, $\triangle AOB$ is a right triangle with legs of length $OB = 8$ and $OA = 15$. Therefore, $AB = 17$. From here, we can solve the problem in several ways. Here are a couple:



Solution 1: Use similar triangles. Lots of right angles usually means there are similar triangles. Here, we have

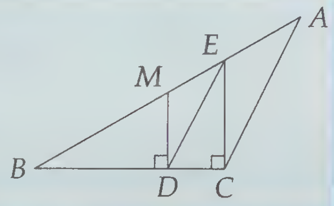
$$\triangle AOB \sim \triangle AEO \sim \triangle OEB.$$

Therefore, $OE/OA = OB/AB$, so $OE = (OA)(OB)/AB = 120/17$.

Solution 2: Use area. Since $[AOB] = (AO)(OB)/2 = (OE)(AB)/2$, we have $OE = (AO)(OB)/AB = 120/17$. (Yeah, I like the area approach better, too.) □

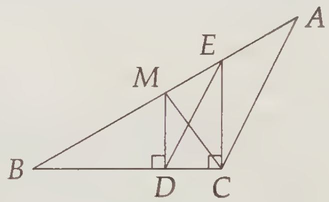
Concept: When in doubt, build right triangles. One very common way to do so is to draw radii to points of tangency. Nearly always try this when you have circles and tangents in a problem.

Problem 19.2: In obtuse triangle ABC shown, $AM = MB$, $\overline{MD} \perp \overline{BC}$, and $\overline{EC} \perp \overline{BC}$. If the area of $\triangle ABC$ is 24, what is the area of $\triangle BED$? *(Source: AMC 12)*



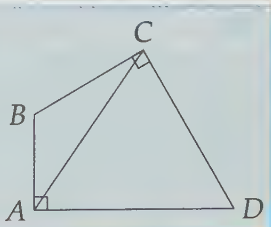
Solution for Problem 19.2: We have a problem about area, and we unfortunately can't find anything about $[BDE]$ easily. We do, however, know how to relate $[AMC]$ and $[BMC]$ to $[ABC]$. Therefore, we connect M to C. (We might also have thought to do this because $\overline{CM}$ is a median, and we know something about medians.)

Since $\overline{CM}$ is a median of $\triangle ABC$, we have $[BMC] = [AMC] = 12$. Now we look for ways to relate either of these to what we want, $[BED]$. Triangles $\triangle BED$ and $\triangle BMC$ overlap. When we remove $\triangle BDM$ from both, we are left with two triangles that share side $\overline{DM}$: $\triangle EDM$ and $\triangle CDM$. Since $\overline{DM} \parallel \overline{EC}$, the altitudes to side $\overline{DM}$ of these triangles are the same. Therefore, $[EDM] = [CDM]$, which means $[BED] = [BMC] = 12$. □



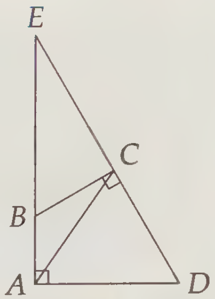
Concept: Connecting points that are originally not connected in a diagram can be extremely useful! This doesn't mean you should connect everything in your diagram immediately, however. Look for segments to draw that will be helpful, particularly those connecting important points, or those that form segments, triangles, or angles you know something about immediately.

Problem 19.3: In triangle ABC, $\angle ABC = 120^\circ$, $AB = 3$ and $BC = 4$. If lines perpendicular to $\overline{AB}$ at A and to $\overline{BC}$ at C meet at D, then find CD. *(Source: AMC 12)*



Solution for Problem 19.3: We start by noting that $\angle D = 360^\circ - 120^\circ - 90^\circ - 90^\circ = 60^\circ$. Next, we might think to draw $\overline{BD}$, since that will give us a couple right triangles. However, we only have one side of those two right triangles, and we don't know anything about the acute angles of them. We'd like to make the 120° and 60° angles useful. This gets us thinking about 30-60-90 triangles.

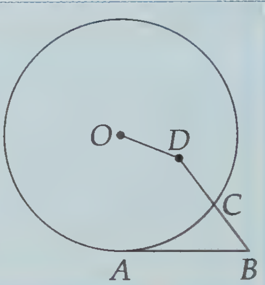
With a right angle at $\angle BAD$ and a 60° angle at $\angle D$, we see that extending $\overline{DC}$ past C and $\overline{AB}$ past B to meet at point E will give us a 30-60-90 triangle. Moreover, since $\angle CBE = 180^\circ - 120^\circ = 60^\circ$ and $\angle BCE = 90^\circ$, $\triangle CBE$ is also a 30-60-90 triangle. Now we can solve the problem.



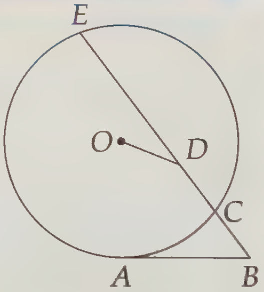
From 30-60-90 $\triangle CBE$ we have $BE = 2BC = 8$ and $EC = BC\sqrt{3} = 4\sqrt{3}$. From $\triangle DAE$ we have $DE = AE(2/\sqrt{3}) = (AB + BE)(2/\sqrt{3}) = 22\sqrt{3}/3$. Therefore, $CD = ED - EC = 10\sqrt{3}/3$. □

Concept: 60°, 30°, and even 120° angles are often good clues to build 30-60-90 triangles by dropping altitudes or extending segments.

Problem 19.4: $\overline{AB}$ is tangent at A to the circle with center O, point D is interior to the circle, and $\overline{DB}$ intersects the circle at C. If $BC = CD = 3$, $OD = 2$, and $AB = 6$, then find the radius of the circle. *(Source: AMC 12)*



Solution for Problem 19.4: We have a problem with segment lengths and a circle, so we think of Power of a Point. We have a tangent, but only part of a secant. We continue $\overline{BD}$ past D until it hits the circle at E. We do this not only because it lets us use Power of a Point, but because $\overline{BD}$ seems to end rather abruptly in the middle of the circle. Segments that seem to end suddenly in a diagram (i.e. ones that, if continued, will hit an important circle or segment) are often candidates to be extended. The power of point B gives us



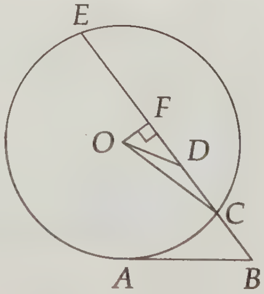
$$(BC)(BE) = BA^2,$$

and substitution gives

$$3(3 + CE) = 36.$$

Therefore, $CE = 9$. Here are a couple ways we can finish:

Solution 1: Build right triangles. We know the length of chord $\overline{CE}$ and we'd like to find the length of a radius. We draw radius $\overline{OC}$ and drop a perpendicular from O to $\overline{CE}$. This gives us a couple right triangles in $\triangle ODF$ and $\triangle OCF$. Since $\overline{OF}$ is part of a radius that is perpendicular to chord $\overline{CE}$, it bisects $\overline{CE}$. Therefore, $CF = 9/2$ and $DF = CF - CD = 3/2$. So,

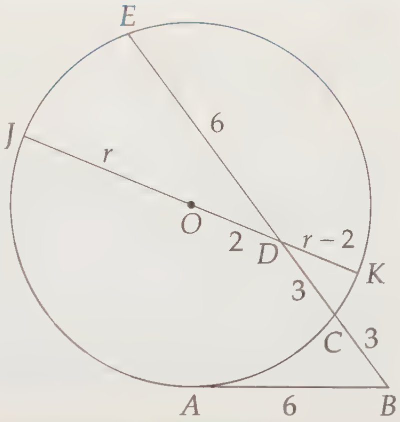


$$OF = \sqrt{OD^2 - DF^2} = \frac{\sqrt{7}}{2}.$$

Finally, we have

$$OC = \sqrt{OF^2 + FC^2} = \sqrt{\frac{7}{4} + \frac{81}{4}} = \sqrt{22}.$$

Solution 2: Keep doing what worked. Extending a line that suddenly stopped worked once, so try it again! $\overline{OD}$ stops rather suddenly. So, we extend it in both directions to hit the circle at J and K, as shown. Now, we have two intersecting chords, so we can use Power of a Point! The power of point D gives



$$(JD)(DK) = (CD)(DE).$$

If we let the radius of the circle be r, we have

$$(r+2)(r-2) = (3)(9-3).$$

Solving for r, we find $r = \sqrt{22}$. □

Concept: Segments that stop suddenly inside figures (particularly triangles, quadrilaterals, or circles) are great candidates to be extended.

Concept: If you successfully use a certain tactic in a problem to get some information, but you still haven't solved the problem, try using that same tactic again in a different way. Maybe it still has more information to give!

Notice that our path to the solution is easy to see in the final diagram of the last solution. This is because the side lengths are labeled.

Concept: Label lengths in your diagram as you find them, even if you have to label them in terms of an important variable.

Exercises

19.1.1 A square is inscribed in a circle of radius 1 as shown at left below. Circles $\odot P$ and $\odot Q$ are the largest circles that can be inscribed in the indicated segments of the circle. The segment joining the centers of circles P and Q intersects the square at A and B. Find AB. *(Source: ARML)* **Hints:** 10

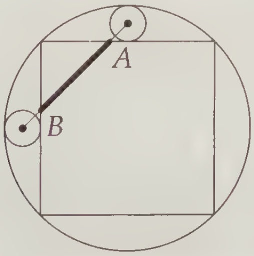


Figure 19.1: Diagram for Problem 19.1.1

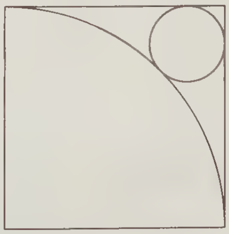


Figure 19.2: Diagram for Problem 19.1.2

19.1.2 In the diagram at right above, a quarter-circle centered at one vertex of the square connects two other vertices of the square. A small circle is tangent to the large circle and to two sides of the square as shown. Each side of the square is 4 units long. What is the radius of the small circle? **Hints:** 19

19.1.3 A round table is pushed into a corner as shown in the diagram. Point A is on the outer edge of the table and is 2 inches from one of the walls. Given that the radius of the table is 37 inches, how many inches is point A from the other wall? *(Source: MATHCOUNTS)* **Hints:** 30

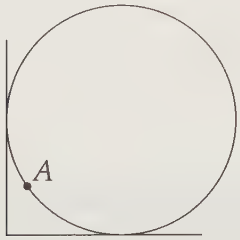


Figure 19.3: Diagram for Problem 19.1.3

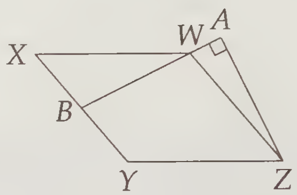


Figure 19.4: Diagram for Problem 19.1.4

19.1.4★ $WXYZ$ is a parallelogram, B is the midpoint of $\overline{XY}$, and A is on $\overleftrightarrow{WB}$ such that $\overline{ZA} \perp \overline{BA}$. Prove $AY = YZ$. **Hints:** 34

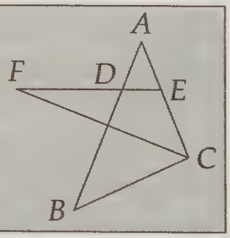
19.2 Assigning Variables

In many problems, finding the answer is not simply a matter of performing routine calculations. Often we need to discover relationships between angles or between lengths. Usually, these relationships will be equations of some sort. One very helpful way to keep track of this information is to assign variables to lengths and/or angles. We can then write equations that are easy to read, and we can use the variables to keep track of the information in our diagram.

As you'll see, we often use this strategy together with our 'extra line' tactic from the last section, particularly when we build right triangles to use the Pythagorean Theorem.

Problems

Problem 19.5: In the figure at right, $AD = AE$, and F is the intersection of $\overrightarrow{ED}$ with the bisector of $\angle C$. If $\angle B = 36°$, how many degrees are there in $\angle CFE$?

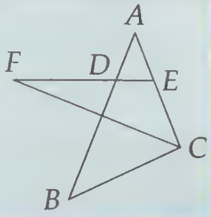


Problem 19.6: Points A, B, C, and D lie on a line, in that order, with $AB = CD$ and $BC = 12$. Point E is not on the line, and $BE = CE = 10$. The perimeter of $\triangle AED$ is twice the perimeter of $\triangle BEC$. Find AE. *(Source: AMC 10)*

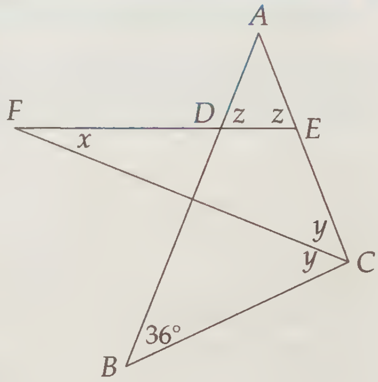
Problem 19.7: $WXYZ$ is a rectangular sheet of paper with $WX = 10$ and $XY = 12$. The paper is folded so that W coincides with the midpoint of $\overline{XY}$. What is the length of the fold?

Variables are great help in angle-chasing problems. As you might guess, the fact that the angles of a triangle add to 180° is our most commonly used equation-forming tool. Try assigning variables and chasing angles in this problem.

Problem 19.5: In the figure at right, $AD = AE$, and F is the intersection of $\overrightarrow{ED}$ with the bisector of $\angle C$. If $\angle B = 36^\circ$, how many degrees are there in $\angle CFE$?



Solution for Problem 19.5: We can't deduce the measures of any more angles directly from $\angle B = 36^\circ$, so we'll have to use relationships among the angles to learn more. We could start off by writing a bunch of equations like:



$$\begin{aligned}
\angle B + \angle ACB + \angle A &= 180^\circ \\
\angle ADE &= \angle AED \\
\angle EDB + \angle DBC + \angle BCE + \angle DEC &= 360^\circ,
\end{aligned}$$

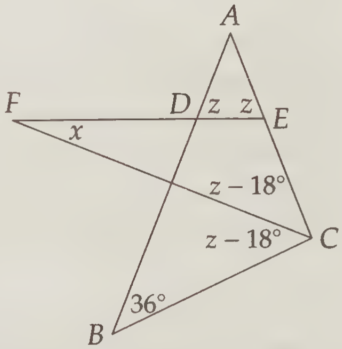
but there are so many different angles that it's hard to keep track of them all. Instead, we assign variables. We assign a variable to what we want, $\angle CFE = x$, and we also pick variables for angles that are easy to relate to other angles in the diagram. We are given that $\triangle ADE$ is isosceles, so we let $\angle ADE = \angle AED = z$. Similarly, our angle bisector gives us $\angle ACF = \angle FCB$, so we call these both y.

Now we look for other angles we can label in terms of x, y, or z. Triangle $\triangle ADE$ gives us $\angle A = 180^\circ - 2z$. Then, $\triangle ABC$ gives us $\angle A + \angle B + \angle ACB = 180^\circ$, so

$$180^\circ - 2z + 36^\circ + 2y = 180^\circ.$$

Therefore, $y = z - 18^\circ$.

Now we can reduce the number of variables in our diagram. If we don't see our path to solution now, we can redraw the diagram using only x and z. Since $\angle FEA$ is an exterior angle of $\triangle FEC$, we have



$$\angle CFE + \angle FCE = \angle DEA.$$

Therefore, $x + z - 18^\circ = z$, so $x = 18^\circ$.

We could also have written $\angle DEC = 180^\circ - z$ in our diagram, then used the sum of the angles of $\triangle FEC$ to solve the problem. □

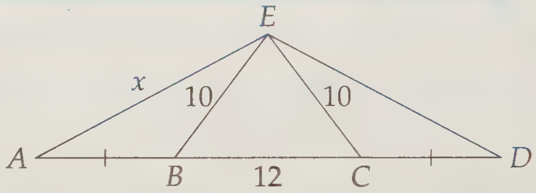
Extra! *The essence of mathematics is in its freedom.*

–Georg Cantor

Concept: Assign variables to important angles and use your mastery of basic geometry to express other angles in terms of those variables. Typically, you should assign variables to angles you seek and to angles you know a lot about first. Once you start chasing angles, reach for all the angle tools you know, most notably the facts you know about triangles, straight lines, parallel lines, and angles that intersect circles.

We can also chase lengths with equations. Our main tools here are the Pythagorean Theorem, similar triangles, and Power of a Point. As we've seen throughout this book, the Pythagorean Theorem is the most heavily used of these three for finding lengths.

Problem 19.6: Points A, B, C, and D lie on a line, in that order, with $AB = CD$ and $BC = 12$. Point E is not on the line, and $BE = CE = 10$. The perimeter of $\triangle AED$ is twice the perimeter of $\triangle BEC$. Find AE. *(Source: AMC 10)*



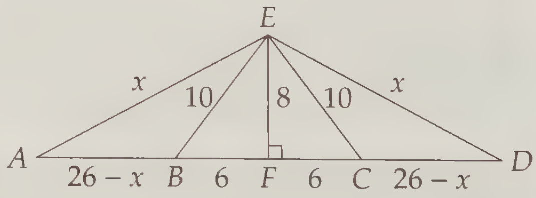
Solution for Problem 19.6: We start with a diagram. We draw everything we are given in the problem. Then we look for more lengths we can find in the diagram and we're almost immediately stuck. So, we start assigning variables. Since we want AE, we let $AE = x$ and look for other sides we can express in terms of x.

We suspect all these equal side lengths ($AB = CD$ and $BE = CE$) will mean $AE = DE$, so we try to prove it. Since $BE = CE$, $\angle EBC = \angle ECB$. Therefore, $\angle EBA = \angle ECD$, so $\triangle ABE \cong \triangle DCE$ by SAS Congruence. So, we have $DE = AE = x$, too.

This isn't enough, so we look for more lengths we can express in terms of x. AB and CD are all we have left. We look back at the problem for information we haven't used yet and see that bit about the perimeter of $\triangle AED$ being twice that of $\triangle BEC$. From this, we have

$$AE + ED + AB + BC + CD = 2(10 + 10 + 12).$$

Substitution gives $2x + 12 + 2AB = 64$, so $AB = 26 - x$.



Adding all this to our diagram, we're still stuck. The isosceles triangles are a clue to build right triangles and use the Pythagorean Theorem. We draw altitude $\overline{EF}$, thereby building right triangles $\triangle EFC$ and $\triangle EFA$. From $\triangle EFC$ we have $EF = 8$. Now we can build an equation for x by applying the Pythagorean Theorem to $\triangle EFA$. $EF^2 + AF^2 = AE^2$, so

$$64 + (32 - x)^2 = x^2.$$

Therefore, $64 = x^2 - (32 - x)^2$. We factor the right side of this equation as the difference of squares to find:

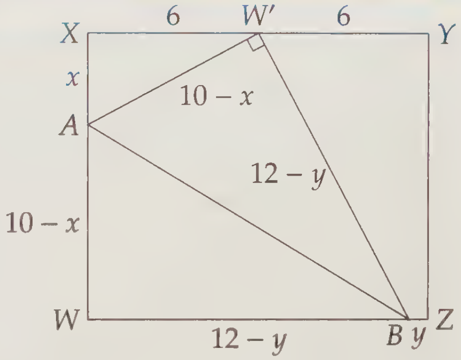
$$64 = [x - (32 - x)][x + (32 - x)] = (2x - 32)(32).$$

Solving for x gives $x = 17$.

Make sure you follow that nifty use of the difference of squares factorization. This manipulation can often save you a few extra steps of nasty algebra (and will reduce careless mistakes). □

Concept: If you need a length in a problem, try building right triangles. Some of the best tools for building right triangles are drawing altitudes of triangles and trapezoids, and drawing radii to points of tangency.

Problem 19.7: $WXYZ$ is a rectangular sheet of paper with $WX = 10$ and $XY = 12$. The paper is folded so that W coincides with the midpoint of $\overline{XY}$. What is the length of the fold?

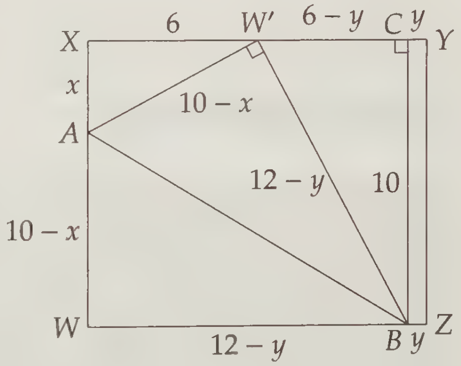


Solution for Problem 19.7: We start with a diagram. Instead of just drawing the folded paper, we also include the paper as it originally was, since we know so much about rectangles. Our diagram shows $WXYZ$, and the fold $\overline{AB}$ that leads to W coinciding with W', the midpoint of $\overline{XY}$. Since $\triangle W'AB$ is just the 'folded over' (i.e. reflected) version of $\triangle WAB$, we have $\triangle W'AB \cong \triangle WAB$. We only initially know $XW' = W'Y = 6$ and $XW = YZ = 10$. We'll have to assign some variables. We let $AX = x$ and $BZ = y$, so we then have $AW = 10 - x$ and $WB = 12 - y$.

We then use our congruent triangles to note that $AW' = AW = 10 - x$ and $W'B = WB = 12 - y$. Right triangle $\triangle XAW'$ gives us an equation for x:

$$x^2 + 6^2 = (10 - x)^2.$$

Solving this equation for x gives $x = 16/5$.



Unfortunately, we don't have a right triangle that quickly gives us y. So, we borrow a tactic from last section and build one. We draw $\overline{BC}$ such that $\overline{BC} \perp \overline{XY}$ as shown. Since $BCYZ$ is a rectangle, we have $CY = y$, $BC = 10$, and $CW' = 6 - y$. We could use the Pythagorean Theorem as before on $\triangle W'CB$. However, we can find the answer a little faster by noting that $\angle CW'B = 180^\circ - \angle AW'B - \angle AW'X = 90^\circ - \angle AW'X = \angle XAW'$. Therefore, right triangles $\triangle W'XA$ and $\triangle BCW'$ are similar. So, we have

$$\frac{AX}{W'C} = \frac{XW'}{BC}.$$

Substitution gives

$$\frac{16/5}{6-y} = \frac{6}{10},$$

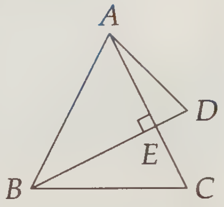
so $y = 2/3$.

Therefore, $WB = BW' = 12 - y = 34/3$. In right triangle $\triangle AWB$, we now have $BW = 34/3$ and $AW = 10 - x = 34/5$, so we can find AB:

$$AB = \sqrt{AW^2 + WB^2} = \sqrt{\left(\frac{34}{5}\right)^2 + \left(\frac{34}{3}\right)^2} = \sqrt{34^2\left(\frac{1}{5}\right)^2 + 34^2\left(\frac{1}{3}\right)^2} = 34\sqrt{\left(\frac{1}{5}\right)^2 + \left(\frac{1}{3}\right)^2} = \frac{34\sqrt{34}}{15}.$$

Notice once again that we use a little algebraic manipulation, in this case factoring out the 34, to simplify our work. □

Exercises

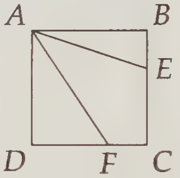


19.2.1 Triangles ABC and ABD are isosceles with $AB = AC = BD$, and $\overline{BD}$ intersects $\overline{AC}$ at E. Given that $\overline{BD} \perp \overline{AC}$, find all possible values of $\angle C + \angle D$. *(Source: AMC 12)*

19.2.2 A point on a circle inscribed in a square is 1 and 2 units from the two closest sides of the square. Find the area of the square. *(Source: HMMT)*

19.2.3 $ABCD$ is a rectangular piece of paper with $AB = 8$ and $AD = 12$. We fold the paper so that B coincides with D. What is the length of the fold? **Hints:** 92

19.2.4★ Square $ABCD$ has side length 1. Point E is chosen on side $\overline{BC}$ so that $AE+EB = \frac{3}{2}$, and point F is chosen on side $\overline{CD}$ so that $\overline{AF}$ bisects $\angle DAE$. Find DF. *(Source: Mandelbrot)* **Hints:** 566, 588, 592

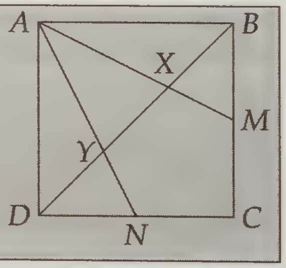


19.3 Proofs

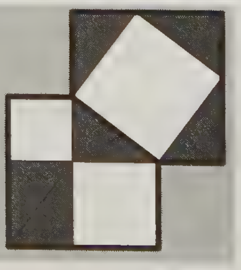
Geometry proofs present a challenge by offering us so many different avenues of exploration. Our task is to narrow down the possible approaches to the ones that are most likely to bear fruit. Before trying the problems in this section, read through Problem 13.10 again, noting particularly how we keep track throughout the problem of What We Know and What We Want. Then, try applying this approach to these problems. We will also use some of our techniques from the first two sections of this chapter, especially adding extra lines.

Problems

Problem 19.8: M and N are the midpoints of sides $\overline{BC}$ and $\overline{CD}$, respectively, of square $ABCD$. $\overline{AM}$ and $\overline{AN}$ meet $\overline{BD}$ at X and Y, as shown. Show that $BX = XY = YD$.

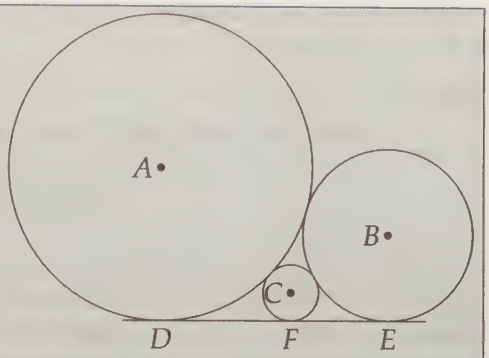


Extra! A 'proof without words' of the Pythagorean Theorem is shown at right. Alexander Bogomolny describes it as an 'unfolded variant' of a proof by abu' l'Hasan Thâbit ibn Qurra Marwân al'Harrani popularized by Monty Phister.



Problem 19.9: $\odot A$ and $\odot B$ are externally tangent. D is on $\odot A$ and E on $\odot B$ such that $\overleftrightarrow{DE}$ is a common external tangent of the two circles. $\odot C$ is tangent to $\odot A$, $\odot B$, and $\overline{DE}$ as shown. Given that a is the radius of $\odot A$, b is the radius of $\odot B$, and c is the radius of $\odot C$, prove that

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}.$$



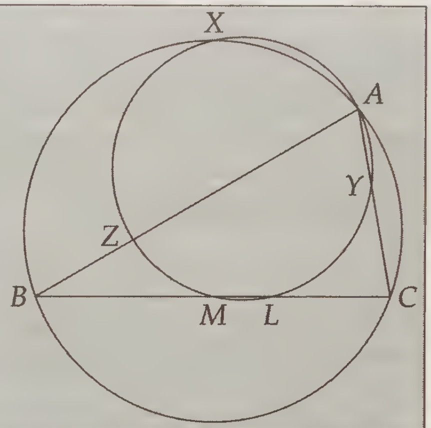
Problem 19.10: In $\triangle ABC$ let L be the foot of the angle bisector from A to $\overline{BC}$, and let M be the midpoint of $\overline{BC}$. We also draw the circumcircle of $\triangle ABC$. Finally, we construct the circle through points A, L, and M. This circle intersects the rest of the diagram in several remarkable ways, so we label points Z, Y, and X where the circle meets $\overline{AB}$, $\overline{AC}$ and the circumcircle of $\triangle ABC$, respectively.

(a) Prove that $BZ = CY$.

(b) Prove that $\triangle XBZ \cong \triangle XCY$.

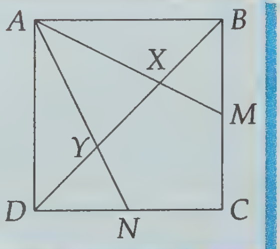
(c) Prove that $\overline{XL}$ is the diameter of the circumcircle of $\triangle ALM$.

(Source: Mandelbrot)



In some of the following problems, we will explore in search of the solution. After some of these problems, we will present brief solutions based on our discoveries. The purpose of the exploration is to give you some insight into how an experienced geometer thinks about problems. The purpose of the solutions following the exploration is to give you a feel for how to write up clear solutions once you find them.

Problem 19.8: M and N are the midpoints of sides $\overline{BC}$ and $\overline{CD}$, respectively, of square $ABCD$. $\overline{AM}$ and $\overline{AN}$ meet $\overline{BD}$ at X and Y, as shown. Show that $BX = XY = YD$.



Solution for Problem 19.8: We make our list of what we know and what we want. We want to prove a statement about lengths, so we start with information we know about lengths.

What We Know	What We Want
$AB = BC = CD = DA$ $BM = MC = CN = DN$	$BX = XY = XD$

It sure seems reasonable that $BX = YD$, but we have to prove it. Our main tool for proving segments equal is congruent triangles, so we look for congruent triangles in our diagram. $\triangle ABX$ and $\triangle ADY$ look congruent, so we go after these two triangles.

Important: Draw precise diagrams. Triangle congruences, parallel lines, collinear points, and much, much more stand out in precise diagrams.

Since $ABCD$ is a square, we have $AD = AB$ and $\angle ADB = \angle ABD = 45°$. We just need another angle to prove congruence. The angles at X and Y don't look easy to work with, but angles $\angle XAB$ and $\angle DAY$ are also parts of right triangles $\triangle BAM$ and $\triangle DAN$. Since $AD = AB$ and $BM = DN$ (both are half the side length of the square), we have $\triangle BAM \cong \triangle DAN$ by LL Congruence (or by SAS Congruence). So, we have $\angle XAB = \angle MAB = \angle NAD = \angle YAD$, which means $\triangle ABX \cong \triangle ADY$ by ASA Congruence.

We have more information for our table now. Most importantly, *we've reduced our problem to just proving* $BX = XY$. We quickly brainstorm for a few other statements that are equivalent to $BX = XY$ and we see that $BY = 2BX$ or $BY = 2DY$ will also give us what we want. Also, if we show $BX = BD/3$ or $XY = BD/3$, we'll be finished.

What We Know	What We Want
$AB = BC = CD = DA$	$BX = XY$
$BM = MC = CN = DN$	$BY = 2BX$
$BX = XD$	$BY = 2DY$
$\triangle ADY \cong \triangle ABX$	$XY = BD/3$
	$BX = BD/3$

This gives us a lot to shoot for. We don't have any more congruent triangles that are interesting to investigate, but the ratios in our 'What We Want' list suggest looking for similar triangles. Parallel lines mean similar triangles. Specifically, we have $\triangle DYN \sim \triangle BYA$ because $\overline{DN} \parallel \overline{AB}$. Seeing $BY = 2DY$ in the 'What We Want' list, we focus on what $\triangle DYN \sim \triangle BYA$ tells us about $\overline{BY}$ and $\overline{DY}$:

$$\frac{DY}{BY} = \frac{DN}{AB} = \frac{CD/2}{AB} = \frac{1}{2}.$$

We've found something we want! Now we retrace our steps to write a nice solution:

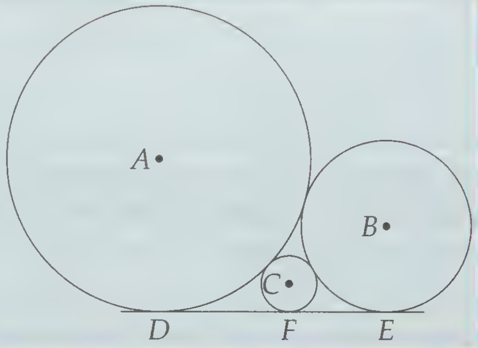
Since $AB = AD$, $\angle ABM = \angle ADN$ and $DN = BM$ (each is half the length of a side of the square), we have $\triangle ADN \cong \triangle ABM$ by SAS Congruence. Therefore, $\angle DAY = \angle DAN = \angle MAB = \angle XAB$. Together with $AB = AD$ and $\angle ABX = \angle YDA$ (since each is 45°), we have $\triangle ABX \cong \triangle ADY$ by ASA Congruence. Therefore, $BX = DY$.

Extra! Josey and Beth are standing 100 feet apart with a 100 foot rope connecting their ankles such that the rope is pulled taut 1 inch above the ground between them. Josey steps one foot towards Beth, so the rope is relaxed along the ground. Is it now possible to lift the center of the rope high enough that a person can walk under it without ducking or moving either of the girls? Is it possible for Josey to lift the rope above her head without her and Beth getting closer (and without her lifting her foot)?

Because $\overline{DN} \parallel \overline{AB}$, we have $\angle NDY = \angle YBA$ and $\angle YAB = \angle YND$, so $\triangle DYN \sim \triangle BYA$. Therefore, $DY/BY = DN/AB = (CD/2)/AB = 1/2$. Therefore, $DY = BY/2$, so $BX = BY/2$ and $XY = BY - BX = BY/2$ also. Thus, $BX = XY = YD$. (See if you can find other solutions!) □

Problem 19.9: $\odot A$ and $\odot B$ are externally tangent. D is on $\odot A$ and E on $\odot B$ such that $\overleftrightarrow{DE}$ is a common external tangent of the two circles. $\odot C$ is tangent to $\odot A$, $\odot B$, and $\overline{DE}$ as shown. Given that a is the radius of $\odot A$, b is the radius of $\odot B$, and c is the radius of $\odot C$, prove that

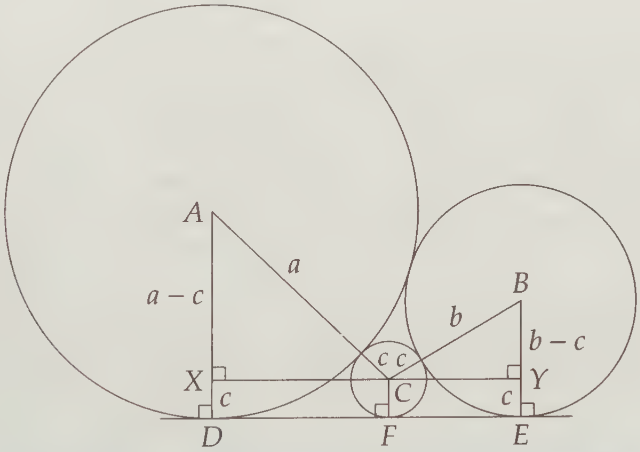
$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}.$$



Solution for Problem 19.9: There's not much working backwards we can do here, but we can at least get rid of the fractions by mutliplying our desired equation by $\sqrt{abc}$. This gives

$$\sqrt{ab} = \sqrt{bc} + \sqrt{ac}.$$

Going forwards, we can use all the information we know about tangent circles and lines. We connect centers and draw radii. The perpendicular lines thus formed and the square roots in our target equation both point in the same direction: the Pythagorean Theorem. In looking for a good right triangle to build, we remember Problem 12.22, in which we built a right triangle to find the length of a common tangent to two circles. Inspired by this, we draw a line through C perpendicular to $\overline{AD}$ and $\overline{BE}$. Rather than listing everything we know in a table, we add it to our diagram:



Concept: A large diagram in which we keep close track of what we discover is a *very* effective way to keep track of What We Know.

Now we have our right triangles. Since $XDFC$ and $CYEF$ are rectangles, we have $XD = CF = YE = c$. Therefore, $AX = a - c$ and $BY = b - c$. Since $AC = a + c$, we can apply the Pythagorean Theorem to $\triangle AXC$

to find

$$\begin{aligned} XC &= \sqrt{AC^2 - AX^2} \\ &= \sqrt{(a+c)^2 - (a-c)^2} \\ &= \sqrt{4ac} \\ &= 2\sqrt{ac}. \end{aligned}$$

Our result is part of the equation we want to prove! Since $XCFD$ is a rectangle, we have $DF = 2\sqrt{ac}$ as well. Therefore, the common external tangent of tangent circles with radii a and c has length $2\sqrt{ac}$. We can apply this result to our other two pairs of tangent circles to find $FE = 2\sqrt{bc}$ and $DE = 2\sqrt{ab}$. Since $DE = DF + FE$, we have

$$2\sqrt{ab} = 2\sqrt{ac} + 2\sqrt{bc}.$$

Dividing this equation by $2\sqrt{abc}$ gives the desired

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}.$$

Notice that we used both of our tactics from earlier in this chapter. We drew many extra lines to build right triangles and we labeled lengths we could find in terms of our variables. As we did so, we found lengths that are in our expression we sought. This brought us right to the solution. □

Concept: Always compare new problems to problems you have already solved. Problem 19.9 is very similar to Problem 12.22, in which we found the length of a common tangent. Thinking of this common external tangent problem gave us a quick path to the solution of our new problem.

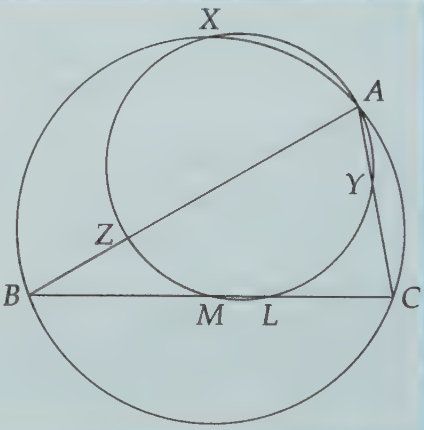
Problem 19.10: In $\triangle ABC$ let L be the foot of the angle bisector from A to $\overline{BC}$, and let M be the midpoint of $\overline{BC}$. We also draw the circumcircle of $\triangle ABC$. Finally, we construct the circle through points A, L, and M. This circle intersects the rest of the diagram in several remarkable ways, so we label points Z, Y, and X where the circle meets $\overline{AB}$, $\overline{AC}$ and the circumcircle of $\triangle ABC$, respectively.

(a) Prove that $BZ = CY$.

(b) Prove that $\triangle XBZ \cong \triangle XCY$.

(c) Prove that $\overline{XL}$ is the diameter of the circumcircle of $\triangle ALM$.

(Source: Mandelbrot)



Solution for Problem 19.10:

(a) We don't have an obvious pair of congruent triangles to go after to show that $BZ = CY$. (We suspect that we'll need $BZ = CY$ to tackle the triangles in the second part, so we don't go after those triangles immediately.) We pull out our other length tools that this diagram invites us to use. The circles suggest Power of a Point. Point B gives us $(BZ)(BA) = (BM)(BL)$ and C

gives $(CY)(CA) = (CL)(CM)$. These equations include our target lengths! The Angle Bisector Theorem gives us $AB/BL = AC/CL$, which has one of our target lengths, and the midpoint gives us $BM = CM$. We put all this information together:

What We Know	What We Want
$(BZ)(BA) = (BM)(BL)$ $(CY)(CA) = (CL)(CM)$ $AB/BL = AC/CL$ $BM = CM$	$BZ = CY$

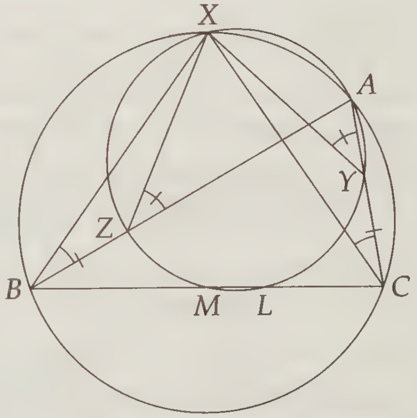
Solving our first two equations in the What We Know column for BZ and CY gives us $BZ = (BM)(BL)/(BA)$ and $CY = (CL)(CM)/(CA)$. We'd like the expressions on the right sides of these equations to be equal:

What We Know	What We Want
$(BZ)(BA) = (BM)(BL)$ $(CY)(CA) = (CL)(CM)$ $AB/BL = AC/CL$ $BM = CM$	$BZ = CY$ $(BM)(BL)/(BA) = (CL)(CM)/(CA)$

Since $CM = BM$, our second line of what we want reduces to $BL/BA = CL/CA$. This is just the reciprocal of the third line in What We Know! Now we have a path from What We Know to What We Want, so we can write our proof:

> From the power of point B, we have $(BZ)(BA) = (BM)(BL)$, or $BZ = (BM)(BL)/(BA)$. The Angle Bisector Theorem applied to bisector $\overline{AL}$ of $\triangle ABC$ gives $BL/BA = CL/CA$, so $BZ = (BM)(BL/BA) = (BM)(CL/CA)$. $BM = CM$ because M is the midpoint of $\overline{BC}$, so $BZ = (CM)(CL/CA)$. Finally, the power of point C gives us $(CY)(CA) = (CL)(CM)$, from which we have $CY = (CM)(CL/CA) = BZ$.

(b) The previous part gave us $BZ = CY$, so if we find two pairs of equal corresponding angles in $\triangle BZX$ and $\triangle CYX$, we will prove the triangles are congruent.



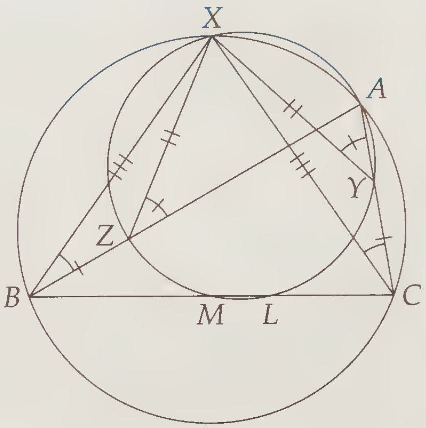
We have circles, so we look for angles that are inscribed in the same arc. Angles $\angle XBZ$ and $\angle XCY$ are the same as $\angle XBA$ and $\angle XCA$, respectively. These two angles are both inscribed in $\widehat{XA}$ of the larger circle, so we have $\angle XBZ = \angle XCY$. Unfortunately, we can't do the same for any other pair of angles of our triangles. We therefore start chasing angles, looking for a pair of equal angles that we might relate to the angles in our triangles.

We continue looking for equal inscribed angles, and find $\angle AZX = \angle AYX$ since both are inscribed in $\widehat{AX}$ of the smaller circle. We can quickly relate each to angles in our triangles:

$$\angle BZX = 180^\circ - \angle AZX = 180^\circ - \angle AYX = \angle XYC.$$

Therefore, by ASA Congruence, we have $\triangle XBZ \cong \triangle XCY$. Notice how marking equal angles and sides as we find them makes the congruent triangles stand out.

(c) We don't have a lot of tools for proving a segment is a diameter of a circle. One possible approach is to show that one of the angles inscribed in either $\widehat{XYL}$ or $\widehat{XZL}$ is a right angle, thus making the arc $180°$. We consider the labeled points on the small circle, looking for a point to be the vertex of our sought-after right angle.



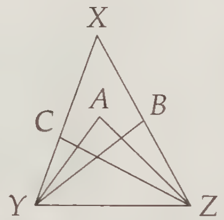
Marking the equal sides given by $\triangle XBZ \cong \triangle XCY$ helps guide us. Our congruent triangles tell us $BX = CX$, so $\triangle XBC$ is isosceles. Therefore, midpoint M of $\overline{BC}$ is also the foot of the altitude from X to $\overline{BC}$. Since $\angle XMC$ is a right angle, $\angle XML = \angle XMC$, and $\angle XML$ is inscribed in $\widehat{XYL}$, we have $\widehat{XYL} = 180°$. Therefore, $\overline{XL}$ is a diameter of the small circle.

As an Exercise, you'll find another approach to this part.

□

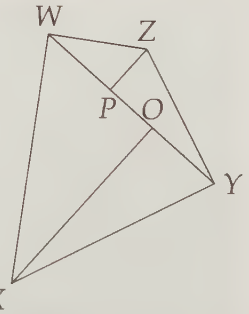
Exercises

19.3.1 In triangle ABC, let the bisector of angle BAC intersect $\overline{BC}$ at D and the circumcircle of triangle ABC at E. Prove that $AE \cdot DE = BE^2$. **Hints:** 260



19.3.2 In the diagram, $\overline{ZA}$ bisects $\angle BZC$, and $\overline{YA}$ bisects $\angle BYX$. Prove that $\angle YCZ + \angle YBZ = 2\angle YAZ$. **Hints:** 134

19.3.3 Prove $\angle LZX = \angle LYX$ in Problem 19.10 without using the fact that $\overline{XL}$ is a diameter of the small circle. Use this to find an alternate proof that $\overline{XL}$ is a diameter of the circumcircle of $\triangle ALM$.

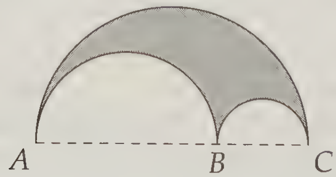


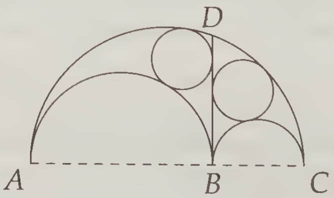
19.3.4 Quadrilateral $WXYZ$ has right angles at $\angle W$ and $\angle Y$ and an acute angle at $\angle X$. Altitudes are dropped from X and Z to diagonal $\overline{WY}$, meeting $\overline{WY}$ at O and P as shown. Prove that $WO = PY$.

19.3.5★ Let A, B, and C be points on a line, in that order. We draw semicircles on segments $\overline{BC}$, $\overline{AC}$, and $\overline{AB}$. Then the figure they enclose is called an *arbelos*, which is shaded in the diagram below at left. The arbelos has many interesting properties, two of which we prove in this problem.

(a) Take point D on arc $\widehat{AC}$ such that $\overline{DB} \perp \overline{AC}$. Prove that the arbelos has the same area as the circle with diameter $\overline{BD}$. **Hints:** 340

(b)★ Line $\overline{BD}$ from the previous part divides the arbelos into two parts. Inscribe a circle in each part. Prove that the two circles have equal radius. **Hints:** 295, 330, 284, 221





19.4 Summary

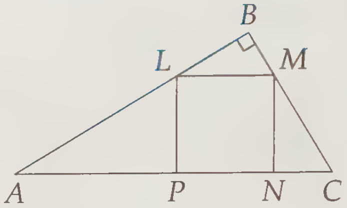
This entire chapter is about problem solving strategies. Here are several of the most useful approaches we used to solve problems throughout this book.

- When in doubt, build right triangles. One very common way to do so is to draw radii to points of tangency. Nearly always try this when you have circles and tangents in a problem. Altitudes of triangles and trapezoids also often produce useful right triangles.
- Connecting points that are originally not connected in a diagram can be extremely useful! This doesn't mean you should connect everything in your diagram immediately, however. Look for segments to draw that will be helpful, particularly those connecting important points, or those that form segments, triangles, or angles you know something about immediately.
- 60°, 30°, and even 120° angles are good signs to try to build 30-60-90 triangles by dropping altitudes or extending segments.
- Consider extending segments that stop suddenly inside figures such as triangles, quadrilaterals, or circles.
- If you successfully use a certain tactic in a problem to get some information, but you still haven't solved the problem, try using that same tactic again in a different way. Maybe it still has more information to give!
- Label lengths as you find them, even if you have to label them in terms of an important variable.
- Assign variables to important angles and use your mastery of basic geometry to express other angles in terms of those variables. Typically, you should first assign variables to angles you seek and to angles you know a lot about.
- Once you start chasing angles, reach for all the angle tools you know, most notably the facts you know about triangles, straight lines, parallel lines, and angles that intersect circles.
- When trying to find a geometric proof, keep careful track of what you know and what you want.
- Draw large, precise diagrams – triangle congruences, parallel lines, collinear points, and much, much more stand out more clearly in large, precise diagrams than in small, scribbled ones.
- Compare new problems to problems you have already solved.

Challenge Problems

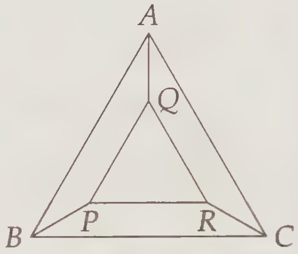
19.11 $\triangle XYZ$ is a right triangle with right angle $\angle X$. P is on $\overline{XZ}$. The triangle is folded over $\overline{YP}$ so that point X lands on side $\overline{YZ}$. Given that $XY = 6$ and $PY = PZ$, find, with proof, $\angle XZY$.

19.12 Square $LMNP$ is inscribed in right triangle ABC as shown. Given $PN = 6$, compute $(AP)(NC)$. *(Source: ARML)*



19.13 Four circles of radius 2 are arranged so that each is tangent to two others, and their centers are the vertices of a square of side length 4. A small circle is inside the square such that it is tangent to the four circles. What is the radius of the small circle?

19.14 $\triangle ABC$ and $\triangle PQR$ are equilateral in the diagram at right. $\overline{AQ}$, $\overline{RC}$, and $\overline{BP}$ are congruent and bisect their respective angles. The four interior regions $AQPB$, $AQRC$, $BPRC$, and QPR all have the same area. Given that $AB = 20$, what is PB? *(Source: MATHCOUNTS)*

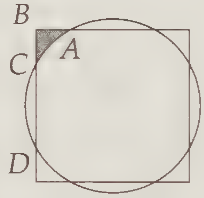


19.15 Point P is inside rectangle $ABCD$. Prove that $AP^2 + CP^2 = BP^2 + DP^2$. **Hints:** 344, 380

19.16 $WXYZ$ is a parallelogram. M is the midpoint of $\overline{WZ}$ and O is the midpoint of $\overline{YZ}$. $\overline{MO}$ and $\overline{XZ}$ meet at N. Find ZN/NX.

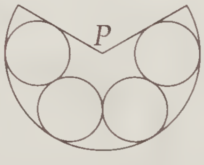
19.17 In $\triangle ABC$, $\angle BAC = 60°$, $\angle ACB = 45°$, and D is on $\overline{BC}$ such that $AD = 18$ and $\angle BAD = \angle CAD$. Find the area of $\triangle ABC$. **Hints:** 411, 443, 473

19.18 In the diagram at right, the circle and the square have the same center. If the area of shaded region ABC equals the area bounded by $\overline{CD}$ and minor arc $\overset{\frown}{CD}$, compute the ratio of the side of the square to the radius of the circle. *(Source: ARML)*



19.19 In quadrilateral $ABCD$, $\overline{AB} \parallel \overline{CD}$. $\overline{AC}$ and $\overline{BD}$ meet at E. Points M and N are the midpoints of $\overline{AE}$ and $\overline{DE}$, respectively. $\overline{BM}$ and $\overline{BE}$ trisect $\angle ABC$, and $\overline{CE}$ and $\overline{CN}$ trisect $\angle BCD$. Prove that $ABCD$ is a rectangle. **Hints:** 501, 539

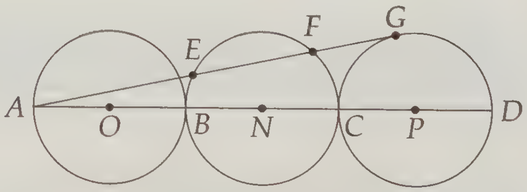
19.20 Four congruent circles are tangent to each other and tangent to the edges of a sector as shown. If the straight edges are joined to form a right circular cone with vertex at P, the radius of the base would be 2/3 the slant height of the cone. Compute the ratio of the radius of the sector to the radius of each circle. *(Source: ARML)* **Hints:** 562, 535



19.21 $ABCD$ is a square. Parallel lines m, n, and p pass through vertices A, B, and C, respectively. The distance between m and n is 7 units, and the distance between n and p is 9 units. Find the number of square units in the area of square $ABCD$. *(Source: MATHCOUNTS)* **Hints:** 502, 468

19.22 Chords $\overline{XY}$ and $\overline{TU}$ of $\odot O$ bisect each other. Furthermore, $XY = TU$. Prove that the two chords meet at the center of the circle.

19.23 In the figure, points B and C lie on line segment $\overline{AD}$, and $\overline{AB}$, $\overline{BC}$, and $\overline{CD}$ are diameters of circles O, N, and P, respectively. Circles O, N, and P all have radius 15, and the line AG is tangent to circle P at G. If $\overline{AG}$ intersects circle N at points E and F, then find EF. *(Source: AHSME)* **Hints:** 502, 418



19.24 Find the volume of a sphere that is inscribed in a regular octahedron with side length 12. **Hints:** 337, 279, 242

19.25 In the diagram at left below, O is the center of the circle. Find the area of the shaded region given that $\angle AOC = 120°$ and $OM = MC = 6$.

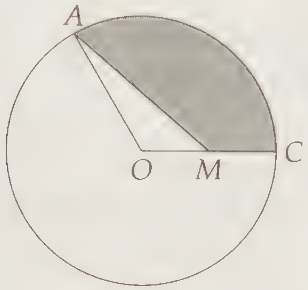


Figure 19.5: Diagram for Problem 19.25

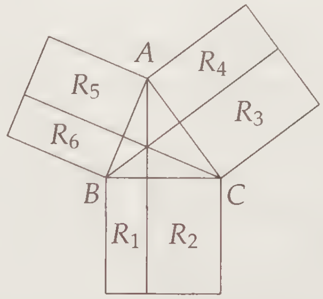


Figure 19.6: Diagram for Problem 19.26

19.26 Squares are erected externally on the sides of $\triangle ABC$ as shown at right above. The altitudes from vertices A, B, and C are drawn and extended to divide these squares into rectangles. Let R_i, for $1 \le i \le 6$, be the areas of the rectangles as shown. Prove that $R_2 = R_3$, $R_4 = R_5$, and $R_6 = R_1$. What does this result say if $\angle ACB = 90°$? **Hints:** 496, 446

19.27 Shown at left below is right rectangular prism $ABCDEFGH$. Prove that $\triangle ACF$ is acute. **Hints:** 396, 365

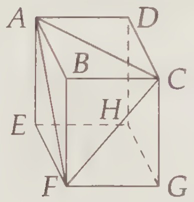


Figure 19.7: Diagram for Problem 19.27

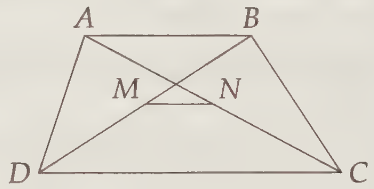
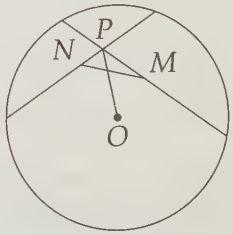


Figure 19.8: Diagram for Problem 19.28

19.28 M and N are the midpoints of diagonals $\overline{BD}$ and $\overline{AC}$, respectively, of trapezoid $ABCD$ at right above. $\overline{AB} \parallel \overline{CD}$, $AB = 8$, and $MN = 6$.

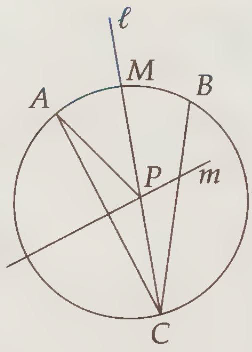
(a) Prove that $\overline{MN} \parallel \overline{CD}$. (Find a rigorous proof – this is one of those 'obvious' facts that requires a careful proof.) **Hints:** 333

(b) Find CD. **Hints:** 146, 492



19.29★ A circle in the plane has center O. Two chords, with midpoints M and N, intersect at P. Prove that $MN \le OP$. When does equality occur? **Hints:** 206, 239

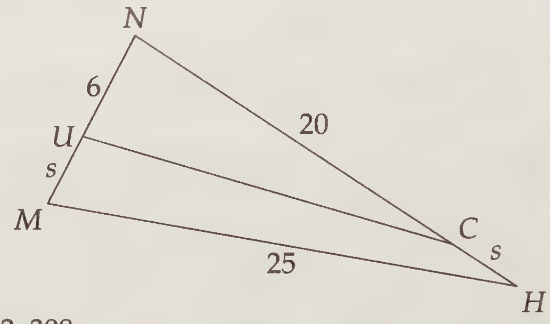
19.30★ Let R and S be points on the sides $\overline{BC}$ and $\overline{AC}$, respectively, of triangle ABC, and let P be the intersection of $\overline{AR}$ and $\overline{BS}$. Determine the area of triangle ABC if the areas of triangles APS, APB, and BPR are 5, 6, and 7, respectively. *(Source: USAMTS)* **Hints:** 286, 251



19.31 In the diagram to the right, points A and B are on $\odot O$. Point C is on major arc $\widehat{AB}$. Line ℓ is the angle bisector of $\angle ACB$ and line m is the perpendicular bisector of $\overline{AC}$. Lines ℓ and m meet at point P, point M is the midpoint of minor arc $\widehat{AB}$, and minor arc $\widehat{AB}$ has measure θ. Prove that $\angle APM = \theta/2$. *(Source: Mandelbrot)* **Hints:** 313

19.32 Sphere $\mathcal{Z}$ is tangent to all 6 edges of regular tetrahedron $ABCD$. Given that each edge of the tetrahedron has length 12, find the radius of sphere $\mathcal{Z}$. **Hints:** 205, 230, 268

19.33★ $\triangle ABC$ is an equilateral triangle with side length 6. Prove that for any point P we choose inside $\triangle ABC$, the sum of the distances from P to the sides of $\triangle ABC$ is the same. **Hints:** 110



19.34★ U and C are points on the sides of $\triangle MNH$ such that $MU = s$, $UN = 6$, $NC = 20$, $CH = s$, $HM = 25$. If $\triangle UNC$ and $MUCH$ have equal areas, what is s? *(Source: HMMT)* **Hints:** 275

19.35★ Let C_1 and C_2 be two circles that are externally tangent, and let their centers be O_1 and O_2, respectively. Let P be a point on C_1 and let Q be a point on C_2 such that $\overline{PQ}$ is a common external tangent to circles C_1 and C_2, and let M be the midpoint of $\overline{PQ}$. Prove that $\angle O_1MO_2 = 90°$. **Hints:** 283, 308

19.36★ Four balls of radius 1 are all tangent to each other. What is the radius of the smallest sphere that encloses them? **Hints:** 334

19.37★ Diagonals $\overline{AC}$ and $\overline{BD}$ of regular heptagon $ABCDEFG$ meet at X. Prove that $AB + AX = AD$. **Hints:** 359